Mathematics and Physics for Agriculture

Land Sciences

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Mathematics and Physics for Agriculture (A36581)

Short Title: Maths Physics
Faculty Science and Computing
Department Land Sciences
Cluster Mathematics and Physics
Author TWOODCOCK
Credits 5 **Level:** Introductory

Description of Module / Aims

This module will introduce the student to linear and non-linear algebra. There is an emphasis throughout this module on the role of mathematics in the sciences, including real world applications. This module also introduces the student to some fundamental principles of physics and their application to the physical world/ environment. The module contains a practical component where the student can develop measurement, analysis and report-writing skills.

Programmes

`MATH-0032`	BSc in Agriculture (`WD_SAGUL_D`)	1 / 2 / M

Indicative Content

- Mathematic Fundamentals: review of arithmetic operations; fractions; significant figures; decimal places; percentages; ratios; prefixes, scientific notation; vectors and scalars
- Introduction to Algebra: simplifying algebraic expressions; factorisation and algebraic fractions
- Transposition and solution of equations: rearranging a formula; solving linear equations; solving simultaneous equations; solving quadratic equations
- Introduction to trigonometry: degrees; sine rule and cosine rule
- Introduction to measurement, SI units and the need for standards
- Mechanics: motion in one dimension, Newton's laws of motion; work, energy and power; mechanical energy, total energy and momentum; simple machines and levers
- Properties of matter: properties & structure of matter; density and relative density; pressure, atmospheric pressure and hydraulic pressure; temperature and temperature scales
- Electric circuits; resistance, current and Ohm's Law; simple DC circuits and electrical safety
- Where possible, practical Agricultural applicaions and examples will be used when teaching the above content

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use and manipulate routine algebraic and numerical calculations.
2. Construct linear and non-linear equations of varying complexity.
3. Use and apply trigonometric functions in the context of applications to physics.
4. Describe and explain some fundamental principles of mechanics and properties of matter.
5. Apply problem-solving skills to analyse simple electric circuit problems.
6. Complete experiments, record and analyse experimental data and present results in the form of a report.

Learning and Teaching Methods

- Lectures

- Self-directed learning
- Laboratory experiments
- Students are supported with notes, online material and feedback

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	5,6
Lab Report	30%	5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter

40%--49%: Will be able to show a basic knowledge of the key learning outcomes.

50%--59%: As well as the above the student will be able to demonstrate an understanding of some of the complexities of the topics.

60%--69%: In addition, the student will demonstrate more detailed knowledge of the all topics including and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material.

70%--100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learn

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Croft, A. and R. Davison. _Foundation Mathematics_. 7th ed.. Flanders: Pearson, 2020.

Supplementary Material

- "Khan Academy." https://www.khanacademy.org
- Bloomfield, L. _How Things Work: The Physics of Everyday Life_. 6th ed.. London: Wiley, 2015.
- Giancoli, D. _Physics Principles with Applications_. 7th ed.. Flanders: Pearson, 2016.
- O'Regan, D. _Real-World Physics_. Dublin: Folens, 2000.

Requested Resources

- SCIENCE LAB: Physics